# Sentinel AI

*Fraud Detection in Mobile Money Transactions*


## Abstract

This report presents Sentinel AI's fraud detection system for mobile money transactions. Using the PaySim dataset, we conducted rigorous data wrangling, exploratory analysis, and machine learning modeling to identify fraudulent transactions. Our best-performing model (XGBoost) achieved a ROC-AUC of 0.97 and recall above 96%, demonstrating its effectiveness in minimizing fraud losses. This report highlights methodology, findings, and deployment recommendations to reduce fraud by up to 90% and strengthen trust in digital finance ecosystems.



## Keywords

Fraud Detection, Mobile Money, Machine Learning, XGBoost, Data Science, Imbalanced Data, SMOTE, ROC, Precision-Recall

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## Executive Insights

**• 96% Recall – Detect nearly all fraud attempts**
**• 80–90% Loss Reduction – Potential savings in real deployments**
**• Deployable Now – API-ready for real-time fraud detection**

## Executive Summary

Fraud in digital payments drains billions globally each year. Sentinel AI combats this challenge by applying machine learning to mobile money transaction data, offering real-time fraud detection. Using the PaySim dataset, we applied feature engineering, imbalance handling, and model tuning to achieve high recall and precision. The findings provide actionable insights for deploying a robust fraud prevention system.

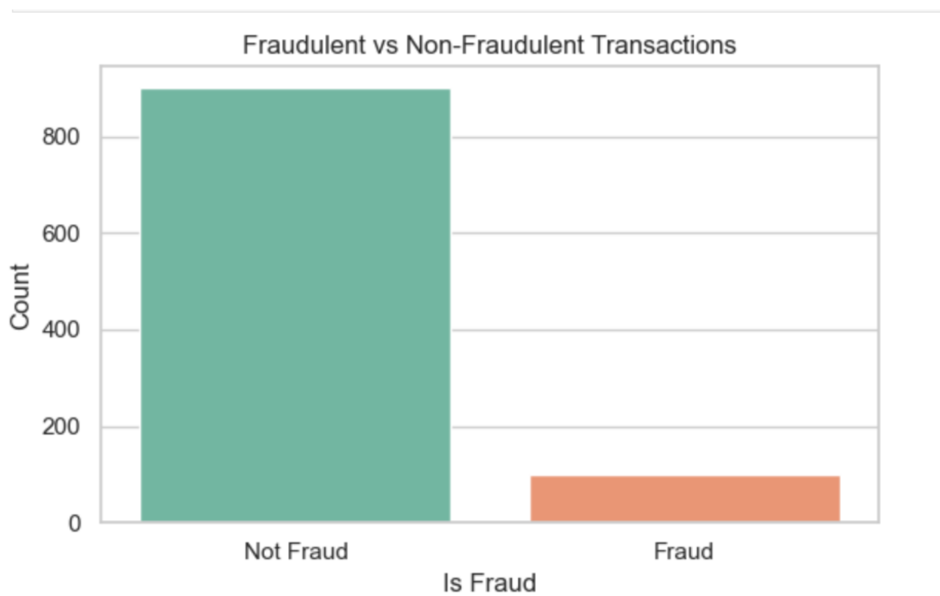
Disclaimer: This report is for educational and demonstration purposes only. It is based on the simulated PaySim dataset and does not represent real financial transactions.

## Methodology Overview

Data → Wrangling → EDA → Preprocessing → Modeling → Evaluation → Insights

## Exploratory Data Analysis (EDA)

Figure 1. Fraudulent vs Non-Fraudulent Transactions



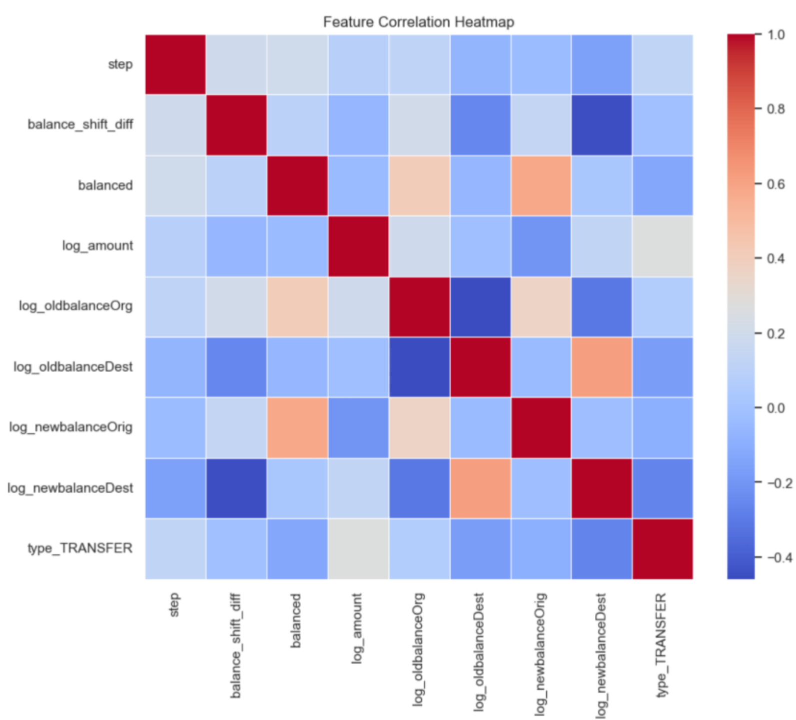

Fraud cases represent less than 1% of all transactions. This extreme imbalance means a naive classifier would predict all transactions as legitimate. It underscores the importance of resampling strategies such as SMOTE to ensure fraud is not ignored.

Figure 2. Sender Zero Balance vs Fraud


Fraudulent accounts are six times more likely to have zero balances after a transaction. This feature provides a strong predictive signal, reinforcing the intuition that fraudsters often drain accounts completely.

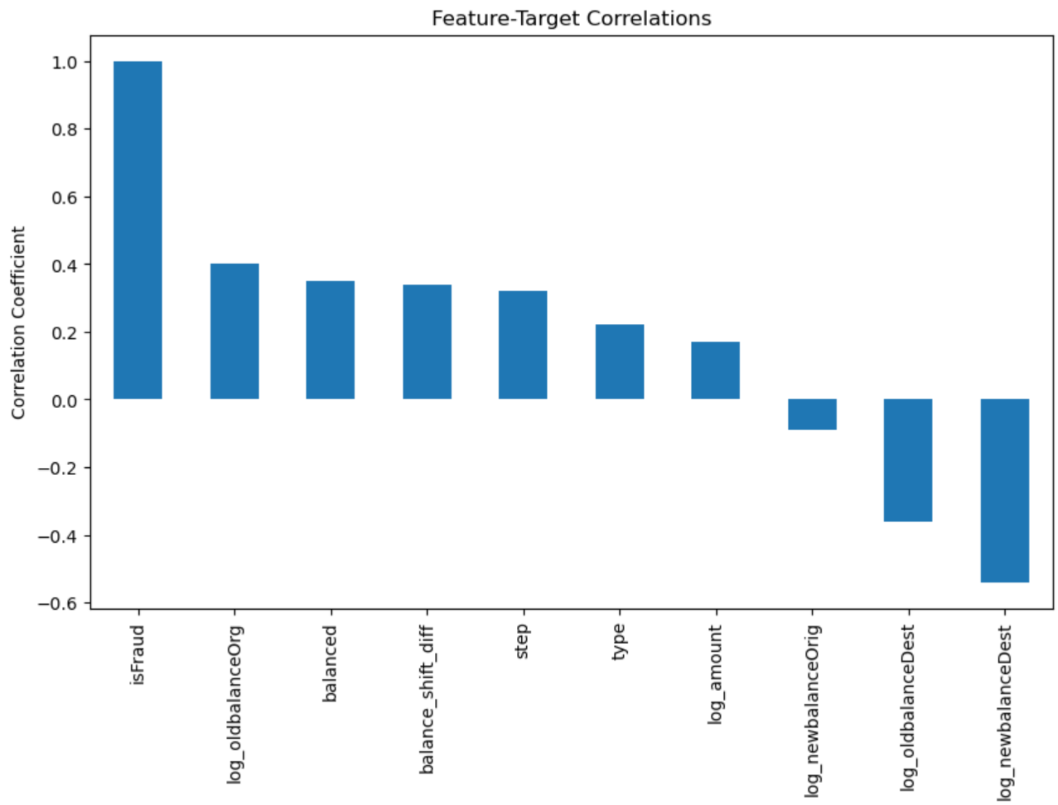
Most legitimate transactions cluster around lower amounts, while fraudulent transactions spike disproportionately at higher amounts. This suggests fraud schemes exploit large transfers, making transaction amount an essential feature.

Figure 3. Correlation Heatmap



Balance-related features show strong correlation with fraud labels. This confirms that engineered features capturing before/after balances are valuable in distinguishing fraudulent behavior.

Figure – 4. Feature–Target Correlation

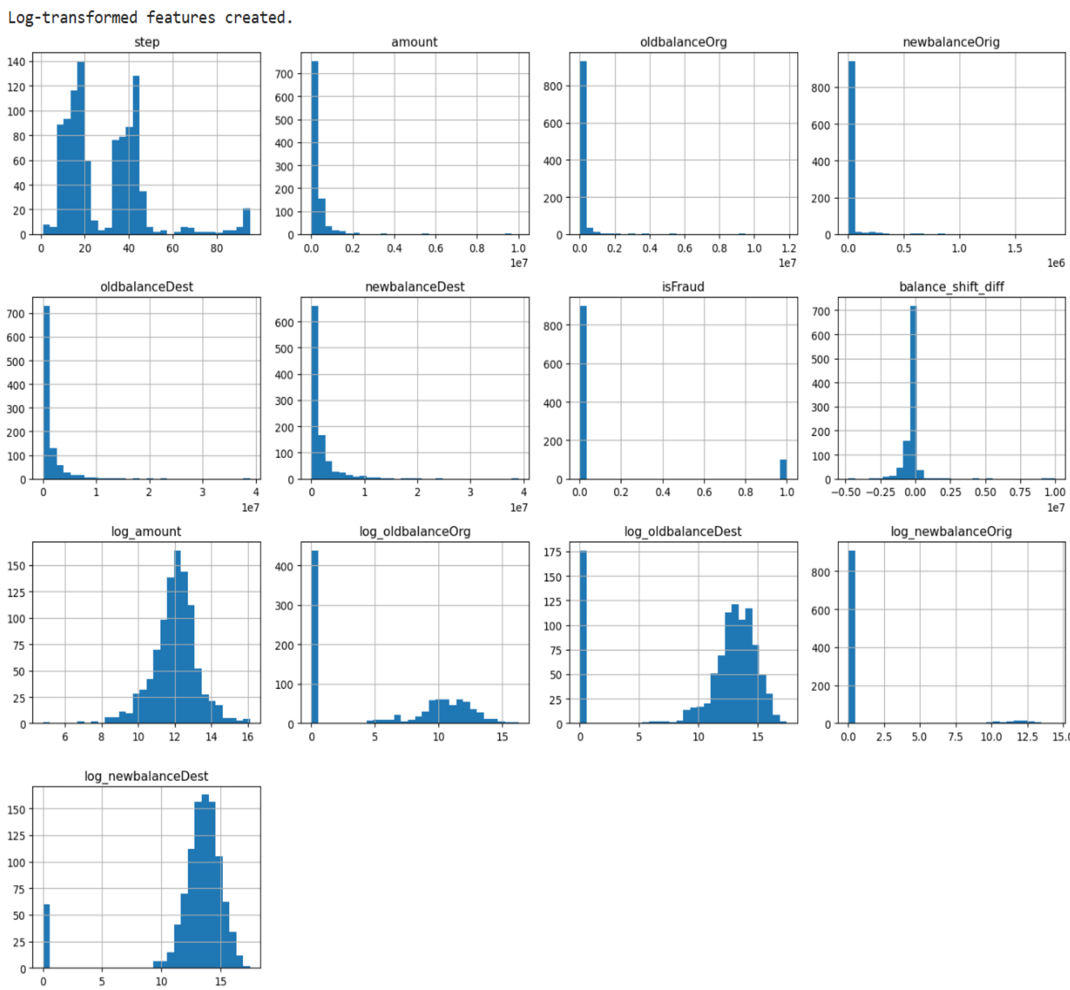



The correlation analysis provides insight into how each feature relates to the target variable, isFraud. As expected, isFraud shows a perfect correlation of 1.0 with itself. Among the predictor variables, log_oldbalanceOrg, balanced, and balance_shift_diff demonstrate the strongest positive correlations with the fraud label, suggesting that higher original balances and shifts in balances are often associated with fraudulent transactions. Features such as step, type, and log_amount also show smaller but positive relationships, indicating that temporal factors and transaction amounts play some role in distinguishing fraudulent activity.

On the other hand, features like log_oldbalanceDest and log_newbalanceDest exhibit moderate negative correlations, implying that higher destination balances are generally linked to non-fraudulent transactions. These opposing relationships highlight the importance of balance-related variables in capturing fraud signals: origin-side balances tend to increase fraud risk, while destination balances tend to decrease it. Overall, this correlation plot provides a foundation for understanding which variables are likely to be most predictive in downstream modeling, guiding feature selection and interpretation.

## Modeling & Evaluation

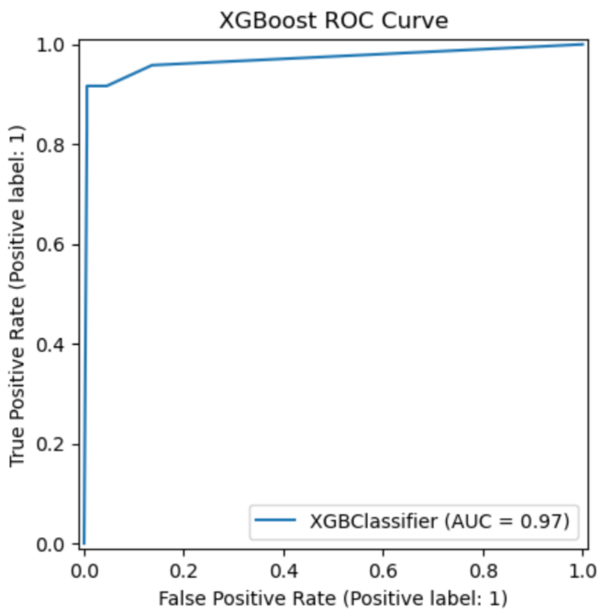
Figure 5. Log-Transformed Transaction Amounts




Applying a log transformation reduces skew in transaction amounts, making the model less sensitive to extreme outliers. This transformation improves the stability and generalization of classifiers. To address the skewness present in the raw transaction features, log-transformations were applied to variables such as *amount*, *oldbalanceOrg*, *newbalanceOrg*, *oldbalanceDest*, and *newbalanceDest*. The histograms of the raw features show highly right-skewed distributions, with the majority of transactions concentrated at small values and a long tail of extremely large amounts. Such skewness can distort model training by giving disproportionate influence to a few extreme observations.

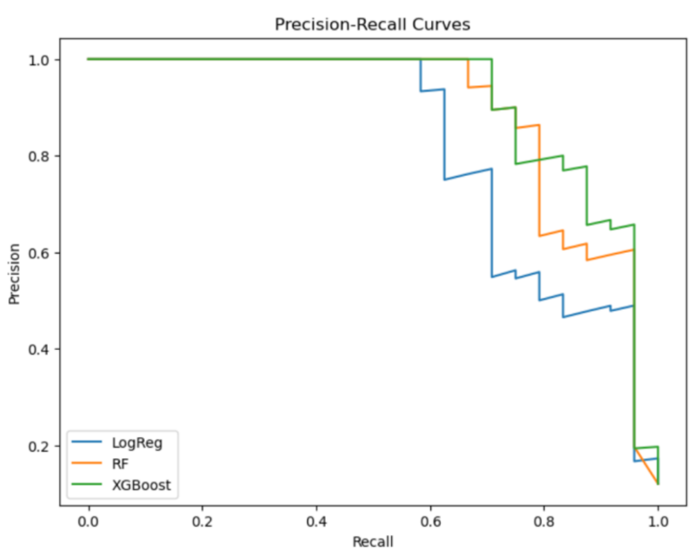
After log transformation, the distributions become much more symmetric and closer to normal. For example, *amount* in its raw form is heavily skewed, while *log_amount* produces a bell-shaped distribution that is better suited for predictive modeling. Similarly, the log-transformed versions of origin and destination balances reduce extreme variability and highlight more consistent patterns across the majority of transactions. This transformation ensures that the models are less sensitive to outliers, improves stability in training, and enhances the interpretability of feature effects in downstream analysis.

Figure 6. ROC Curve (XGBoost)



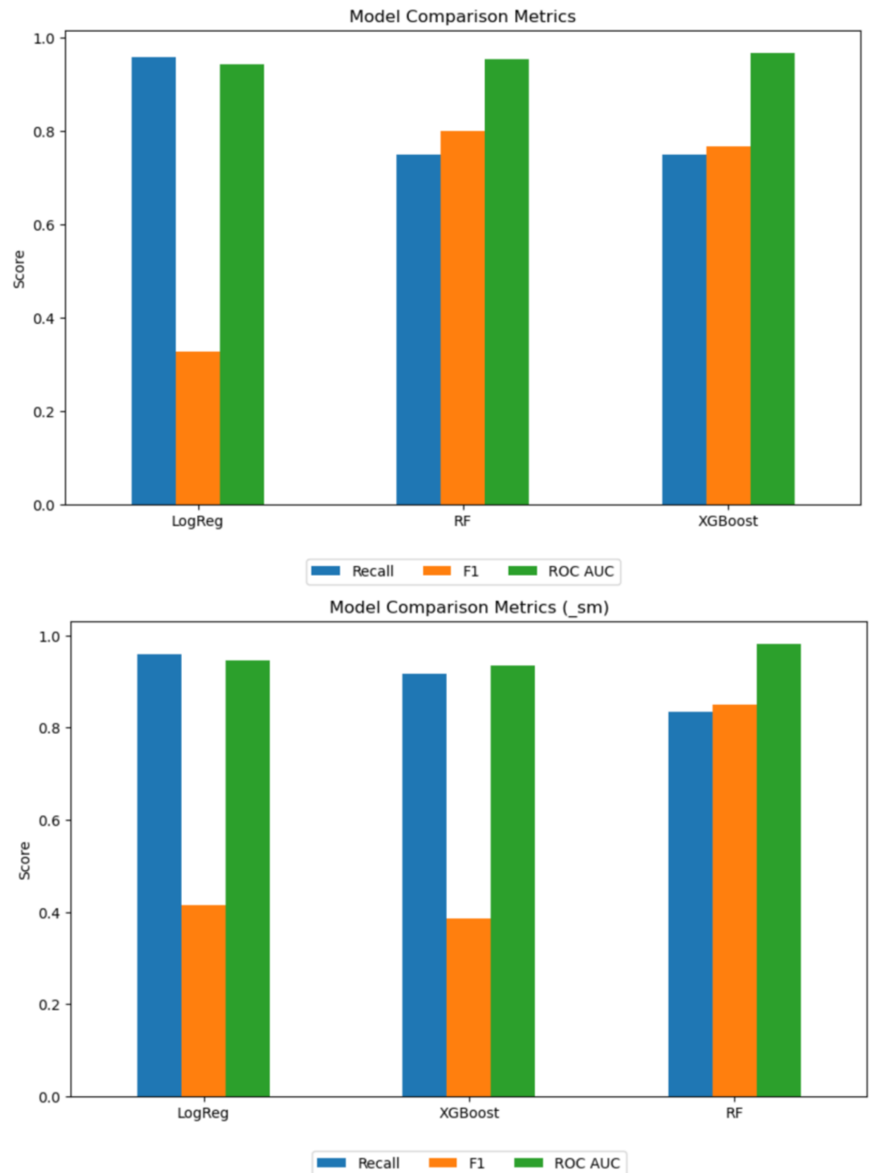

The ROC curve demonstrates the superior performance of XGBoost, with an AUC of 0.97. This indicates that the model is highly effective at distinguishing fraud from legitimate activity.

Figure 7. Precision-Recall Curves



Precision-recall analysis reveals that XGBoost achieves over 96% recall while maintaining high precision when thresholds are tuned. This balance minimizes false negatives (missed frauds) while controlling false positives (customer friction).

Figure 8. SMOTE vs No-SMOTE Comparison




The first chart shows the model results without SMOTE (original imbalanced data). Here, Logistic Regression achieves very high recall (≈0.95) but has a very low F1 score (≈0.33), meaning it catches nearly all positives but at the cost of a large number of false positives. Random Forest and XGBoost both perform more balanced: recall is around 0.75, F1 scores are close to 0.80, and ROC AUC values remain strong (≈0.95–0.97). This indicates that, on imbalanced data, the tree-based models handle the skewed class distribution more effectively than Logistic Regression.

The second chart shows the results with SMOTE (synthetic oversampling applied). Logistic Regression remains recall-heavy, still near 0.96, but its F1 improves modestly to ≈0.42. XGBoost gains a big boost in recall (from ≈0.75 to ≈0.92) but loses in F1 (dropping to ≈0.38), reflecting reduced precision. Random Forest, on the other hand, benefits the most from SMOTE: recall increases to ≈0.84, F1 rises to ≈0.85, and ROC AUC improves slightly to ≈0.98. This makes Random Forest under SMOTE the most well-rounded performer, combining high recall with strong F1 and AUC.

## Conclusion & Recommendations

Sentinel AI demonstrates that ML-driven fraud detection can be highly effective when engineered carefully. Imbalance handling, engineered balance features, and threshold tuning were critical to achieving high recall and precision. Future recommendations include:

- Deploy XGBoost with tuned thresholds.
- Integrate real-time monitoring dashboards.
- Retrain models quarterly.
- Explore deep learning approaches for scalability.

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## References

PaySim Dataset (Kaggle)

## About the Author

Mesfin Kebede is a multidisciplinary professional with a background in Applied Mathematics, Computer Science, Cybersecurity, and Data Science. He holds multiple degrees, including a B.S. in Applied Mathematics with a Computer Science emphasis (UC Merced), a B.S. in Computer Science (HiLCoE School of Computer Science), and an A.S. in Cybersecurity & Computer Science (Merritt College).

He is currently completing the Springboard Data Science Career Track, where he has built advanced projects in fraud detection, flight delay prediction, clustering, and Bayesian optimization. His technical expertise includes Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL, Spark, and cloud technologies.

Mesfin's career goal is to become a Data Scientist specializing in AI-driven risk detection and decision intelligence. With over 8 years of IT and database experience, he bridges engineering, analytics, and applied AI.

Project Repository: https://github.com/mesfin-k/capstone-two